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3D CAPACITANCE STRUCTURE AND MANUFACTURE METHOD THEREOF

Abstract

A 3D capacitance structure, including multiple sub-capacitance stack structures, multiple first electrode contacts, and multiple second electrode contacts, is provided. Each sub-capacitance stack structure includes a dielectric stack, a lower electrode structure, and an upper electrode structure. The dielectric stack includes a stack of dielectric layers and etching stop layers. The lower electrode structure includes a lower electrode plate disposed in a lower dielectric layer and multiple lower electrode extended portions passing upward from the lower electrode plate through a lower etching stop layer to an intermediate dielectric layer. The upper electrode structure includes an upper electrode plate disposed in an upper dielectric layer and multiple upper electrode extended portions extending downward from the upper electrode plate to the lower etching stop layer. The first electrode contacts and the second electrode contacts are respectively connected to lower and upper electrode plates of two adjacent layers in the sub-capacitance stack structures.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of Taiwan application serial no. 113108848, filed on Mar. 11, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure relates to a capacitance structure, and in particular to a 3D capacitance structure and a manufacturing method thereof.

Description of Related Art

[0003] Due to the technical bottlenecks caused by the continuous miniaturization of chips, the industry has shifted from improving the manufacturing process to increase the number of transistors in a single silicon wafer area to improving overall performance through complex system-level chip design with relatively controllable costs. “Chiplet” has received much attention to implement higher transistor density and performance at a relatively low cost.

[0004] The chiplet is to split many elements originally included in a single chip into multiple small units, and then individually strengthen the function, redesign, and remanufacture to form a system chip through advanced packaging technology.

[0005] However, a single capacitor in the chiplet is limited to being manufactured on a silicon substrate, but cannot be made into a capacitor that is too high in capacitance, so the capacitance is also limited and cannot be increased.

SUMMARY

[0006] A 3D capacitance structure of the disclosure includes a substrate, multiple sub-capacitance stack structures, multiple first electrode contacts, and multiple second electrode contacts. The sub-capacitance stack structures are disposed on the substrate. Each sub-capacitance stack structure includes a dielectric stack, a lower electrode structure, and an upper electrode structure. The dielectric stack includes a stack of a lower dielectric layer, a lower etching stop layer, an intermediate dielectric layer, an upper etching stop layer, and an upper dielectric layer. The lower electrode structure includes a lower electrode plate disposed in the lower dielectric layer and multiple lower electrode extended portions passing upward from the lower electrode plate through the lower etching stop layer to the intermediate dielectric layer. The upper electrode structure includes an upper electrode plate disposed in the upper dielectric layer and multiple upper electrode extended portions passing downward from the upper electrode plate through the upper etching stop layer and the intermediate dielectric layer to the lower etching stop layer. The first electrode contacts respectively connect the lower electrode plates of two adjacent layers in the sub-capacitance stack structures. The second electrode contacts respectively connect the upper electrode plates of two adjacent layers in the sub-capacitance stack structures.

[0007] A manufacturing method of a 3D capacitance structure of the disclosure includes the steps of (a) forming a first lower electrode plate on a substrate; (b) forming a first dielectric stack on the first lower electrode plate, wherein the first dielectric stack includes a stack of a lower dielectric layer, an etching stop layer, and an upper dielectric layer; (c) forming multiple first lower electrode extended portions in the first dielectric stack, wherein the first lower electrode extended portions

pass upward from the first lower electrode plate through the etching stop layer to the upper dielectric layer; (d) forming a second dielectric stack, wherein a structure of the second dielectric stack is the same as the first dielectric stack; (e) forming multiple first upper electrode extended portions in the second dielectric stack, wherein the first upper electrode extended portions pass through an etching stop layer of the second dielectric stack to extend to the etching stop layer of the first dielectric stack; (f) forming a first upper electrode plate in the upper dielectric layer of the second dielectric stack, wherein the first upper electrode plate is connected to the first upper electrode extended portions; (g) forming a third dielectric stack, wherein a structure of the third dielectric stack is the same as the first dielectric stack; (h) forming a first contact opening passing through the third dielectric stack, the second dielectric stack, and the first dielectric stack until the first lower electrode plate is exposed; (i) forming a lower electrode plate trench in an upper dielectric layer of the third dielectric stack, wherein the lower electrode plate trench and the first contact opening form a first dual damascene opening; (j) forming a conductor material in the first dual damascene opening to simultaneously form a first electrode contact and a second lower electrode plate; (k) repeating (b) to (d) to form a fourth dielectric stack, multiple second lower electrode extended portions, and a fifth dielectric stack; (l) forming a second contact opening passing through the fifth dielectric stack, the fourth dielectric stack, and the third dielectric stack until the first upper electrode plate is exposed; (m) forming multiple extended portion openings passing through the fifth dielectric stack and an upper dielectric layer of the fourth dielectric stack until an etching stop layer of the fourth dielectric stack is exposed; (n) forming an upper electrode plate trench in an upper dielectric layer of the fifth dielectric stack, wherein the upper electrode plate trench, the extended portion openings, and the second contact opening form a second dual damascene opening; (o) forming a conductor material in the second dual damascene opening to simultaneously form a second electrode contact, multiple second upper electrode extended portions, and a second upper electrode plate; and (p) repeating (g) to (o) at least once.
[0008] In order for the features and advantages of the disclosure to be more comprehensible, the following specific embodiments are described in detail in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0009] FIG. **1** is a schematic cross-sectional view of a 3D capacitance structure according to a first embodiment of the disclosure.
[0010] FIG. **2** is a schematic cross-sectional view of a 3D capacitance structure along a line I-I′ of FIG. **1**.
[0011] FIG. **3** is a schematic cross-sectional view of another 3D capacitance structure along a line I-I′ of FIG. **1**.
[0012] FIG. **4** is a schematic cross-sectional view of a 3D capacitance structure according to a second embodiment of the disclosure.
[0013] FIG. **5**A to FIG. **5**S are schematic cross-sectional views of a manufacturing process of a 3D capacitance structure according to a third embodiment of the disclosure.
DESCRIPTION OF THE EMBODIMENTS
[0014] FIG. **1** is a schematic cross-sectional view of a 3D capacitance structure **10** according to a first embodiment of the disclosure. Please refer to FIG. **1**. The 3D capacitance structure **10** of the embodiment includes a substrate **100**, multiple sub-capacitance stack structures SC, multiple first electrode contacts E**1**, and multiple second electrode contacts E**2**. The sub-capacitance stack structure SC is disposed on the substrate **100**, wherein each sub-capacitance stack structure SC includes a dielectric stack DS, a lower electrode structure LE, and an upper electrode structure UE. The dielectric stack DS includes a stack of a lower dielectric layer **102**, a lower etching stop layer

104, an intermediate dielectric layer **106**, an upper etching stop layer **108**, and an upper dielectric layer **110**. In an embodiment, the lower dielectric layer **102**, the intermediate dielectric layer **106**, and the upper dielectric layer **110** include silicon oxide, and the lower etching stop layer **104** and the upper etching stop layer **108** include silicon nitride. However, the disclosure is not limited thereto. The lower electrode structure LE includes a lower electrode plate **112** disposed in the lower dielectric layer **102** and multiple lower electrode extended portions **114** passing upward from the lower electrode plate **112** through the lower etching stop layer **104** to the intermediate dielectric layer **106**. The upper electrode structure UE includes an upper electrode plate **116** disposed in the upper dielectric layer **110** and multiple upper electrode extended portions **118** passing downward from the upper electrode plate **116** through the upper etching stop layer **108** and the intermediate dielectric layer **106** to the lower etching stop layer **104**. In other words, the 3D capacitance structure **10** has three sub-capacitance stack structures SC. However, the disclosure is not limited thereto. In another embodiment, the number of stacks of the sub-capacitance stack structures SC may be higher.

[0015] In FIG. **1**, the extending directions of the lower electrode extended portion **114** and the upper electrode extended portion **118** are perpendicular to the surface direction of the substrate **100**. The surface directions of the lower electrode plate **112** and the upper electrode plate **116** are perpendicular to the extending directions of the first electrode contact E**1** and the second electrode contact E**2**.

[0016] Please continue to refer to FIG. **1**. The first electrode contacts E**1** respectively connect the lower electrode plates **112** of two adjacent layers in the sub-capacitance stack structures SC, wherein each first electrode contact E**1** directly contacts the lower electrode plate **112** of the adjacent upper layer and the lower electrode plate **112** of the adjacent lower layer. The second electrode contacts E**2** respectively connect the upper electrode plates **116** of two adjacent layers in the sub-capacitance stack structures SC, wherein each second electrode contact E**2** directly contacts the upper electrode plate **116** of the adjacent upper layer and the upper electrode plate **116** of the adjacent lower layer. Since the heights of the first electrode contact E**1** and the second electrode contact E**2** connecting the upper and lower sub-capacitance stack structures SC are fixed at approximately one dielectric stack DS, the stacked 3D capacitance structure **10** can be formed without other connection structures. In addition, the upper etching stop layer **108** and the lower etching stop layer **104** in the dielectric stack DS are conducive to controlling the thickness of each layer. For example, the size (for example, the height) of the upper electrode extended portion **118** may be controlled through the lower etching stop layer **104**, so the most suitable stack can be obtained. In addition, another etching stop layer **120** may be added between the upper and lower sub-capacitance stack structures SC as a stop layer during a process of forming the lower electrode plate **112**.

[0017] In order to connect circuits of the 3D capacitance structure **10** to external or other elements, an interconnection structure may be formed on the substrate **100**. For example, the interconnection structure has an insulation layer **122** covering the entire 3D capacitance structure **10**, a first contact window C**1** connected to the lower electrode plate **112**, a second contact window C**2** connected to the upper electrode plate **116**, a first metal layer **124** located on the insulation layer **122** and connected to the first contact window C**1**, and a second metal layer **126** located on the insulation layer **122** and connected to the second contact window C**2**. However, the disclosure is not limited thereto. In another embodiment, the interconnection structure may include multiple first contact windows C**1** and multiple second contact windows C**2**, and the arrangement positions thereof may be changed according to requirements.

[0018] FIG. **2** is a schematic cross-sectional view of the 3D capacitance structure **10** along a line I-I′ of FIG. **1**. FIG. **1** shows a cross-section of the x-z plane, and FIG. **2** shows a cross-section of the y-z plane, so FIG. **2** only shows the dielectric stack DS, the upper electrode plate **116**, the lower electrode plate **112**, and the lower electrode extended portion **114** in the sub-capacitance stack structure SC, and a dotted line portion is the position of the upper electrode extended portion **118**. Therefore, the lower electrode extended portion **114** and the upper electrode extended portion **118** of the first embodiment may be strip structures staggered along the x direction. However, the disclosure is not limited thereto.

[0019] FIG. **3** is a schematic cross-sectional view of another 3D capacitance structure **10** along a line I-I′ of FIG. **1**. FIG. **3** also shows a cross-section of the y-z plane and shows that the lower electrode extended portion **114** and the upper electrode extended portion **118** are columnar structures staggered along the x direction and the y direction. Since an area between the lower electrode extended portion **114** and the upper electrode extended portion **118** of FIG. **3** is greater than the structure of FIG. **2**, FIG. **3** should have a greater capacitance than FIG. **2** in the case where other structures are all the same.

[0020] FIG. **4** is a schematic cross-sectional view of a 3D capacitance structure **40** according to a second embodiment of the disclosure, wherein the same reference numerals as those in the first embodiment are used to represent the same or similar parts and components, and for the relevant contents of the same or similar parts and components, reference may also be made to the contents of the first embodiment, which will not be described again.

[0021] Please refer to FIG. **4**. The 3D capacitance structure **40** of the second embodiment is basically the same as the 3D capacitance structure **10** of the first embodiment. The difference is that the 3D capacitance structure **40** has a total of eight sub-capacitance stack structures SC. In order to ensure that the structure of the first contact window C**1** is stable and does not collapse, the eight sub-capacitance stack structures SC may be divided into two groups, one first contact window C**1** is provided for every 4 lower electrode plates **112**, and the first metal layer **124** is connected by connecting the first contact windows C**1** in series. Since the sizes of the second electrode contact E**2** and the second contact window C**2** between the upper electrode plates **116** are smaller (lower), the original designs may be maintained. However, the disclosure is not limited thereto. In another embodiment, the connection design between the upper electrode plates **116** may be swapped with the connection design between the lower electrode plates **112**.

[0022] FIG. **5**A to FIG. **5**S are schematic cross-sectional views of a manufacturing process of a 3D capacitance structure according to a third embodiment of the disclosure. Please refer to FIG. **5**A. The manufacturing method of the embodiment may first form a first lower electrode plate **502** on a substrate **500** using a manner such as electroplating (step (a)), and then form a first dielectric stack DS**1** on the first lower electrode plate **502** (step (b)). The first dielectric stack DS**1** includes a stack of a lower dielectric layer LD, an etching stop layer SL, and an upper dielectric layer UD. Then, multiple extended portion openings **504** may be formed in the first dielectric stack DS**1** using a photolithographic etching process. Since there is an etching selectivity ratio between the etching stop layer SL and the upper dielectric layer UD and the lower dielectric layer LD in the first dielectric stack DS**1**, the etching process of forming the extended portion openings **504** use different etching gases to etch different materials.

[0023] After that, please refer to FIG. **5**B. The extended portion opening **504** is filled with a conductor material M**1** using a manner such as electroplating to form multiple first lower electrode extended portions **506** in the first dielectric stack DS**1** (step (c)). The first lower electrode extended portion **506** passes upward from the first lower electrode plate **502** through the etching stop layer SL to the upper dielectric layer UD. In an embodiment, the extending direction of the first lower electrode extended portions **506** is perpendicular to the surface direction of the substrate **500**.

[0024] Next, please refer to FIG. **5**C. A planarization process may be first performed using a manner such as chemical mechanical polishing (CMP), the first lower electrode extended portion **506** is kept, and a second dielectric stack DS**2** is then formed (step (d)). The structure of the second dielectric stack DS**2** is the same as the first dielectric stack DS**1**. However, the thickness and the material of each layer of the second dielectric stack DS**2** may be adjusted according to requirements and are not limited to being exactly the same as the first dielectric stack DS**1**.

[0025] Then, please refer to FIG. **5**D. Multiple extended portion openings **508** may be formed in the second dielectric stack DS**2** using a photolithographic etching process until the etching stop layer SL of the first dielectric stack DS**1** is exposed as predetermined formation positions of upper electrode extended portions.

[0026] Next, please refer to FIG. **5**E. An upper electrode plate trench **510** may be formed in the upper dielectric layer UD of the second dielectric stack DS**2** using another photolithographic etching process. The upper electrode plate trench **510** and the extended portion opening **508** form a dual damascene opening.

[0027] Subsequently, please refer to FIG. **5**F. The upper electrode plate trench **510** and the extended portion opening **508** are filled with a conductor material M**2** using a manner such as electroplating to form multiple first upper electrode extended portions **512** in the second dielectric stack DS**2** (step (e)) and form a first upper electrode plate **514** in the upper dielectric layer UD of the second dielectric stack DS**2** (step (f)). Since the embodiment adopts a dual damascene process, the connected first upper electrode plate **514** and first upper electrode extended portions **512** may be simultaneously formed. However, the disclosure is not limited thereto. In another embodiment, the first upper electrode extended portions **512** may be first formed, and the first upper electrode plate **514** is then formed. The first upper electrode extended portion **512** passes through the etching stop layer SL of the second dielectric stack DS**2** to extend to the etching stop layer SL of the first dielectric stack DS**1**. In other words, the size (for example, the height) of the first upper electrode extended portion **512** may be precisely controlled through the etching stop layer SL. In an embodiment, the extending direction of the first upper electrode extended portions **512** is perpendicular to the surface direction of the substrate **500**. In an embodiment, the first lower electrode extended portions **506** and the first upper electrode extended portions **512** are strip structures staggered along the x direction (similar to the structure of FIG. **2**). In another embodiment, the first lower electrode extended portions **506** and the first upper electrode extended portions **512** are columnar structures staggered along the x direction and the y direction (similar to the structure of FIG. **3**).

[0028] Next, please refer to FIG. **5**G. A planarization process such as a manner of CMP may be first performed, the first upper electrode extended portion **512** and the first upper electrode plate **514** are kept, and the third dielectric stack DS**3** is then formed (step (g)). The structure of the third dielectric stack DS**3** is the same as the first dielectric stack DS**1**. However, the thickness and the material of each layer of the third dielectric stack DS**3** may be adjusted according to requirements and are not limited to being exactly the same as the first dielectric stack DS**1**.

[0029] Then, please refer to FIG. **5**H. A first contact opening EO**1** passing through the third dielectric stack DS**3**, the second dielectric stack DS**2**, and the first dielectric stack DS**1** may be formed using a photolithographic etching process until the first electrode lower plate **502** is exposed (step (h)).

[0030] Next, please refer to FIG. **5**I. A lower electrode plate trench **516** is formed in the upper dielectric layer UD of the third dielectric stack DS**3**, and the lower electrode plate trench **516** and the first contact opening EO**1** form a first dual damascene opening (step (i)). Furthermore, in the embodiment, in the step of forming the lower electrode plate trench **516**, the etching stop layer SL of the third dielectric stack DS**3** may be used as a stop layer.

[0031] Then, please refer to FIG. **5**J. A conductor material may be formed in the lower electrode plate trench **516** and the first contact opening EO**1** (the first dual damascene opening) using electroplating to simultaneously form the first electrode contact E**1** and the second lower electrode plate **518** (step (j)). In an example, the extending direction of the first electrode contact E**1** is perpendicular to the surface direction of the first lower electrode plate **502**.

[0032] Next, please refer to FIG. **5**K. The steps from forming the first dielectric stack DS**1** until forming the second dielectric stack DS**2** are repeated to form a fourth dielectric stack DS**4**, multiple second lower electrode extended portions **520**, and a fifth dielectric stack DS**5** (step (k)).

[0033] Then, please refer to FIG. **5**L, a second contact opening EO**2** passing through the fifth dielectric stack DS**5**, the fourth dielectric stack DS**4**, and the third dielectric stack DS**3** is formed until the first upper electrode plate **514** is exposed (step (**1**)).
[0034] Then, please refer to FIG. **5**M, a mask layer **522** may be first formed on the fifth dielectric stack DS**5** to fill the second contact opening EO**2**, wherein the mask layer **522** is, for example, spin-on-carbon (SOC) or photoresist. Then, multiple mask openings **524** may be formed in the mask layer **522** using a photolithography process as predetermined formation positions of upper electrode extended portions.
[0035] Next, please refer to FIG. **5**N. The fifth dielectric stack DS**5** exposed from the mask opening **524** is etched to be removed using the mask layer **522** of FIG. **5**M as an etching mask until the etching stop layer SL of the fourth dielectric stack DS**4** is exposed to form multiple extended portion openings **526** (step (m)). Then, the mask layer **522** of FIG. **5**M is removed.
[0036] Then, please refer to FIG. **5**O. An upper electrode plate trench **528** may be formed in the upper dielectric layer UD of the fifth dielectric stack DS**5** using a photolithography process (step (n)). The upper electrode plate trench **528**, the extended portion openings **526**, and the second contact opening EO**2** form a second dual damascene opening.
[0037] Next, please refer to FIG. **5**P. A conductor material may be formed in the upper electrode plate trench **528**, the extended portion opening **526**, and the second contact opening EO**2** (the second dual damascene opening) using electroplating to simultaneously form the second electrode contact E**2**, multiple second upper electrode extended portions **530**, and a second upper electrode plate **532** (step (o)). In an embodiment, the extending direction of the second electrode contact E**2** is perpendicular to the surface direction of the first upper electrode plate **514**.
[0038] Please refer to FIG. **5**Q again. The steps of FIG. **5**G to the steps of FIG. **5**P are repeated at least once (step (p)) to form a 3D capacitance structure **50**. The disclosure is not limited thereto. The number of repetitions of the foregoing steps may be increased according to requirements to form a 3D capacitance structure with a greater number of stacks.
[0039] Next, please refer to FIG. **5**R. In order to connect circuits of the 3D capacitance structure **50** to external or other elements, an interconnection structure may be formed on the substrate **500**. For example, a space reserved for a contact window next to the 3D capacitance structure **50** is first defined, and an insulation layer **534** covering the entire 3D capacitance structure **50** is then comprehensively formed.
[0040] Then, please refer to FIG. **5**S. The first contact window C**1** connected to the first lower electrode plate **502** and the second contact window C**2** connected to the second upper electrode plate **532** are first formed in the insulation layer **534**, and a first metal layer **536** connected to the first contact window C**1** and a second metal layer **538** connected to the second contact window C**2** are then formed on the insulation layer **534**. However, the disclosure is not limited thereto. In another embodiment, the interconnection structure may include multiple first contact windows C**1** and multiple second contact windows C**2**, and the arrangement positions thereof may be changed according to requirements.
[0041] In summary, in the disclosure, through stacking the sub-capacitance stack structures, and directly connecting the sub-capacitance stack structures with the electrode contacts, the close and overlapping 3D capacitance structures can be formed, thereby increasing the capacitance and enabling the multi-layer structure to be difficult to collapse. Moreover, the etching stop layer is disposed in the sub-capacitance stack structure, so the thickness of each layer can be controlled, thereby obtaining the most suitable stack, that is, obtaining the optimal upper limit configuration according to requirements.
[0042] Although the disclosure has been disclosed in the above embodiments, the embodiments are not intended to limit the disclosure. Persons skilled in the art may make some changes and modifications without departing from the spirit and scope of the disclosure. Therefore, the protection scope of the disclosure shall be defined by the appended claims.

Claims

1. A 3D capacitance structure, comprising: a substrate; a plurality of sub-capacitance stack structures, disposed on the substrate, wherein each of the sub-capacitance stack structures comprises: a dielectric stack, comprising a stack of a lower dielectric layer, a lower etching stop layer, an intermediate dielectric layer, an upper etching stop layer, and an upper dielectric layer; a lower electrode structure, comprising a lower electrode plate disposed in the lower dielectric layer and a plurality of lower electrode extended portions passing upward from the lower electrode plate through the lower etching stop layer to the intermediate dielectric layer; and an upper electrode structure, comprising an upper electrode plate disposed in the upper dielectric layer and a plurality of upper electrode extended portions passing downward from the upper electrode plate through the upper etching stop layer and the intermediate dielectric layer to the lower etching stop layer; a plurality of first electrode contacts, respectively connecting the lower electrode plates of two adjacent layers in the sub-capacitance stack structures; and a plurality of second electrode contacts, respectively connecting the upper electrode plates of two adjacent layers in the sub-capacitance stack structures.
2. The 3D capacitance structure according to claim 1, wherein extending directions of the lower electrode extended portions and the upper electrode extended portions are perpendicular to a surface direction of the substrate.
3. The 3D capacitance structure according to claim 1, wherein surface directions of the lower electrode plate and the upper electrode plate are perpendicular to extending directions of the first electrode contacts and the second electrode contacts.
4. The 3D capacitance structure according to claim 1, wherein the lower electrode extended portions and the upper electrode extended portions are strip structures staggered along an x direction.
5. The 3D capacitance structure according to claim 1, wherein the lower electrode extended portions and the upper electrode extended portions are columnar structures staggered along an x direction and a y direction.
6. The 3D capacitance structure according to claim 1, wherein the lower dielectric layer, the intermediate dielectric layer, and the upper dielectric layer comprise silicon oxide, and the lower etching stop layer and the upper etching stop layer comprise silicon nitride.
7. A manufacturing method of a 3D capacitance structure, comprising: (a) forming a first lower electrode plate on a substrate; (b) forming a first dielectric stack on the first lower electrode plate, wherein the first dielectric stack comprises a stack of a lower dielectric layer, an etching stop layer, and an upper dielectric layer; (c) forming a plurality of first lower electrode extended portions in the first dielectric stack, wherein the first lower electrode extended portions pass upward from the first lower electrode plate through the etching stop layer to the upper dielectric layer; (d) forming a second dielectric stack, wherein a structure of the second dielectric stack is the same as the first dielectric stack; (e) forming a plurality of first upper electrode extended portions in the second dielectric stack, wherein the first upper electrode extended portions pass through an etching stop layer of the second dielectric stack to extend to the etching stop layer of the first dielectric stack; (f) forming a first upper electrode plate in the upper dielectric layer of the second dielectric stack, wherein the first upper electrode plate is connected to the first upper electrode extended portions; (g) forming a third dielectric stack, wherein a structure of the third dielectric stack is the same as the first dielectric stack; (h) forming a first contact opening passing through the third dielectric stack, the second dielectric stack, and the first dielectric stack until the first lower electrode plate is exposed; (i) forming a lower electrode plate trench in an upper dielectric layer of the third dielectric stack, wherein the lower electrode plate trench and the first contact opening form a first dual damascene opening; (j) forming a conductor material in the first dual damascene opening to

simultaneously form a first electrode contact and a second lower electrode plate; (k) repeating (b) to (d) to form a fourth dielectric stack, a plurality of second lower electrode extended portions, and a fifth dielectric stack; (l) forming a second contact opening passing through the fifth dielectric stack, the fourth dielectric stack, and the third dielectric stack until the first upper electrode plate is exposed; (m) forming a plurality of extended portion openings passing through the fifth dielectric stack and an upper dielectric layer of the fourth dielectric stack until an etching stop layer of the fourth dielectric stack is exposed; (n) forming an upper electrode plate trench in an upper dielectric layer of the fifth dielectric stack, wherein the upper electrode plate trench, the extended portion openings, and the second contact opening form a second dual damascene opening; (o) forming a conductor material in the second dual damascene opening to simultaneously form a second electrode contact, a plurality of second upper electrode extended portions, and a second upper electrode plate; and (p) repeating (g) to (o) at least once.

8. The manufacturing method of the 3D capacitance structure according to claim 7, wherein extending directions of the first lower electrode extended portions and the first upper electrode extended portions are perpendicular to a surface direction of the substrate.

9. The manufacturing method of the 3D capacitance structure according to claim 7, wherein surface directions of the first lower electrode plate and the first upper electrode plate are perpendicular to extending directions of the first electrode contact and the second electrode contact.

10. The manufacturing method of the 3D capacitance structure according to claim 7, wherein the first lower electrode extended portions and the first upper electrode extended portions are strip structures staggered along an x direction.

11. The manufacturing method of the 3D capacitance structure according to claim 7, wherein the first lower electrode extended portions and the first upper electrode extended portions are columnar structures staggered along an x direction and a y direction.

12. The manufacturing method of the 3D capacitance structure according to claim 7, wherein in forming the lower electrode plate trench, an etching stop layer of the third dielectric stack is used as a stop layer.